Muhammad Usman Aziz

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OBJECTIVE

AI and Full-Stack Developer specializing in Generative AI, Deep Learning, and Web Technologies. Focused on developing AI-powered applications, ERP automation, improving accessibility, and building scalable real-time systems. Eager to contribute expertise to innovative projects at the intersection of AI, enterprise ERP systems, and modern web development.

EDUCATION

Bachelor of Computer Science, FAST-NUCES, CFD Campus — Sep 2021 - June 2025

SKILLS

Programming Languages	Python, TypeScript, JavaScript, C++, SQL, R, HTML5/CSS3
AI & Machine Learning	Generative AI, LLMs, RAG, Fine-Tuning (LoRA/PEFT), NLP, Deep Learning, PyTorch, TensorFlow, OpenCV, Keras, Transformers, Pandas, NumPy, Scikit-learn
Full-Stack & Web	React.js, TanStack Start, Node.js, Express.js, Frappe / ERPNext, Tailwind CSS, REST APIs, Webhooks, Dexie (IndexedDB), Cloudflare Workers
Mobile Development	React Native, Capacitor, Android Studio, Expo Go
Databases & ERP	MariaDB, MySQL, MongoDB, ERPNext, Google Sheets API, QuickBooks Integration
Developer Tools	Git, GitHub, VS Code, Cursor, Postman, Vite, Jupyter Notebook, Colab, PyCharm

EXPERIENCE

Software Engineer — Oct 2025 - Present
Agrovisions — *Faisalabad, Punjab, Pakistan*

- Developed and customized enterprise ERPNext solutions, automating sales workflows, live stock tracking, and financial invoicing.
- Engineered tamper-proof cryptographic document verification systems and dynamic QR code generation for authentic corporate records.
- Spearheaded automated ETL pipelines to extract, transform, and migrate multi-year legacy QuickBooks Desktop financial data into ERPNext.
- Built and maintained full-stack web applications and REST API integrations connecting external e-commerce platforms with ERPNext.

Computer Science & STEM Instructor — Aug 2019 - Present
The Century Academy of Science Commerce and Arts — *Faisalabad, Punjab, Pakistan*

- 7+ years of experience teaching Computer Science, Mathematics, and Physics to high school and intermediate students.
- Mentored students in core programming fundamentals, algorithmic problem solving, and analytical thinking with excellent academic outcomes.

Artificial Intelligence Intern — July 2024 - Aug 2024
CodeAlpha — *Remote*

- Developed AI-powered Autonomous Robot Delivering in Python.

Data Analyst Intern — July 2024 - Aug 2024
Oasis Infobyte — *Remote*

- Exploratory Data Analysis (EDA) on Retail Sales Data. Used K-means clustering for segmentation.

Machine Learning Intern — Aug 2024 - Sep 2024
GrowIntern — *Remote*

- Applying ML techniques to real-world problems and gaining hands-on experience with model development and deployment.

PROJECTS

Medium Technical Blogs

Deaf-ED (Final Year Project) - (MERN, OpenCV, and LSTM)

- Designed an AI-powered assistive mobile app using Pakistan Sign Language (PSL) and animations to teach Urdu, English, and basic counting, featuring real-time sign assessment via computer vision and parental progress tracking.

Stockly - (TanStack Start, React, TypeScript, Tailwind CSS, Dexie, Capacitor)

- Developed an offline-first inventory, sales, and profit management application with IndexedDB local caching, automated Google Sheets OAuth sync, and cross-platform mobile support via Capacitor.

QuickBooks Desktop Data Extractor - (Python, QBXML SDK, VBScript COM, Web UI)

- Engineered a production-grade SDK extraction utility leveraging QBXML and COM Bridge to export complete company master data and 20+ transaction types into clean, migration-ready JSON with browser Web UI configuration.

QuickBooks to ERPNext Migration Engine - (Frappe, ERPNext, Python, Pandas)

- Engineered an automated ETL migration pipeline and custom Frappe application to parse, map, and import historical QuickBooks accounting, payroll, and inventory records into ERPNext DocTypes.

ERPNext Document Verification Portal - (ERPNext, Frappe, Python, JavaScript, QR Code)

- Built an enterprise document authentication system for ERPNext invoices and certificates featuring tamper-proof cryptographic QR codes and a dedicated public verification portal.

Gemini Web 2 API - (TypeScript, Node.js, REST API)

- Engineered an API gateway exposing web-based Gemini AI capabilities as OpenAI-compatible REST endpoints with session management and multi-turn conversational support.

Medical AI Assistant - (RAG, LLMs, Vector Retrieval)

- Engineered an intelligent clinical assistant utilizing Retrieval-Augmented Generation (RAG) to mimic diagnostic consultations, query reference medical cases, and compile structured patient reports.

Oxigen E-Commerce Platform - (React, Node.js, Express, ERPNext REST API, Tailwind CSS)

- Created a high-performance retail e-commerce platform and admin portal integrated with ERPNext for real-time inventory management, automated order routing, and invoice synchronization.

Shayar AI - (LSTM, GRU, RNN, NLP)

- Built an intelligent Urdu poetry generator utilizing deep recurrent architectures (LSTM/GRU) to craft original rhyming verses from single prompt words with interactive exploration.